Expanding ML-Documentation Standards For Better Security

Cara Ellen Appel
University of Hamburg

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1st Cara Ellen Appel
Universität Hamburg
Hamburg, Germany
ORCID 0009-0006-8113-428X

***Abstract*—This article presents the current state of ML-security and of the documentation of ML-based systems, models and datasets in research and practice based on an extensive review of the existing literature. It shows a generally low awareness of security aspects among ML-practitioners and organizations and an often unstandardized approach to documentation, leading to overall low quality of ML-documentation. Existing standards are not regularly adopted in practice and IT-security aspects are often not included in documentation.**

Due to these factors, there is a clear need for improved security documentation in ML, as one step towards addressing the existing gaps in ML-security. To achieve this, we propose expanding existing documentation standards for ML-documentation to include a security section with specific security relevant information. Implementing this, a novel expanded method of documenting security requirements in ML-documentation is presented, based on the existing Model Cards and Datasheets for Datasets standards, but with the recommendation to adopt these findings in all ML-documentation.

***Index Terms*—machine learning, security, documentation, standards, model cards.**

I. Introduction

Every day, millions of cyberattacks are launched against internet users and organizations [29]. Cyberattacks aim to disrupt or take over control of systems and networks, as well as to steal, exploit or delete data involved in them [23]. The field of IT-security aims to protect the confidentiality, integrity and availability of data and systems [22] [32]. To achieve this, a number of different protections need to be implemented to secure any IT-system [23] [21].

AI has only become well-established in everyday technology usage in the past few years, as publicly accessible AI-technologies have become available directly to consumers. But the idea of creating artificially intelligent systems is an old one [24]. What largely contributed to the widespread application of AI within the last decade is the advent of machine learning, which significantly accelerated AI development and progress [24]. The importance and societal impact of AI continues to grow with it being increasingly used in critical systems [8] [28]. A failure in such a system might result in catastrophic consequences like the loss of human lives [28]. Machine learning is currently at the core of most artificial intelligence systems, and is one of the most developed and widely applicable fields in AI-research [27]. Organizations rely on ML-models more than ever for their operations, which has led to an ever-greater need for robust security to protect ML-systems [20].

Documentation is one area where steps towards better ML-security can be made. Security as a critical aspect of software quality should be discussed in documentation, although in practice this often is not the case, both for ML and traditional software [32]. But existing ML-documentation standards do not include security aspects and these standards often are not used by practitioners [7] [26]. Practitioners similarly do not include security aspects in ML-documentation in practice, due in large part to their lack of security awareness regarding ML-systems [8] [9]. We address the resulting clear need to close the gaps in ML-documentation regarding security in this paper, as well as discussing the larger problems facing ML-security.

II. Methodology

We have conducted an extensive review of the existing literature in three main areas: ML-documentation (both its implementation in practice and the existing standards), ML-security and traditional software documentation. To identify relevant literature, we used Google Scholar due to its broad coverage of recent academic literature and its keyword search functionality. We searched for literature that was published from 2019 onwards, as a preliminary search of commonly cited research in these fields showed that almost all relevant research was conducted since then. We searched for articles with the following keywords, as we identified these as relevant to the topic of this paper: 'ML', 'AI', 'standards', 'security', 'Adversarial Machine Learning', 'model', 'dataset', 'software documentation'. We searched for these terms individually as well as in various combinations like 'ML security documentation'. Furthermore, we looked at the cited research in the articles we found through our search and, when relevant, also included these in our literature review.

Chapters III to VI of this article discuss the state of the research based on our literature review. Chapter VII presents our proposed method of expanding existing ML-documentation standards to include security aspects, with the goal of raising the quality of security documentation for ML. This section draws on the results of the previous chapters. In chapter VIII we discuss challenges in implementing the proposed method, before concluding with recommendations for future work in chapter IX.

The research:

Research Paper: „Expanding ML-Documentation Standards For Better Security“

Building on research from my bachelor thesis

Keywords: machine learning, security, documentation, standards, model cards

The main question:

How can documentation improve security in ML-systems?

What makes ML-security so important?

ML-based systems are gaining importance and increasing their societal impact rapidly

Applied increasingly in critical systems like healthcare, law enforcement and education

Traditional IT-security methods and strategies can’t sufficiently protect ML-systems

Many ML-systems, models and datasets are not securely protected against adversarial attacks

How is RE related to ML-documentation and security?

Documenting requirements is part of RE

Requirements documentation can cover security requirements in-depth

Identifying security requirements in RE can help raise a system's security level

RE can address frequent documentation pitfalls

The main question:

How can documentation improve security in ML-systems?

The path toward an answer:

Literature review regarding these questions:	How can documentation and RE improve ML-security? What is the current state of ML-documentation? What is the current state of ML-security?
Apply the results to ML-documentation:	Expand existing ML-documentation standards to include security aspects Point out challenges and recommendations for future work

What role does documentation play in security?

Highlights from the literature review

Documentation can improve security through:

- Making security-relevant information available
- Embedding security in workflows
- Transparently documenting security measures
- Increasing security awareness

Highlights from the literature review

Documentation can improve security through:

Only if documentation is:
- thorough
- accessible
- up-to-date
- Created

AND only when RE is done and security is covered

What is the current state of ML-documentation?

Highlights from the literature review

Issues within current ML-documentation:

- standards rarely used
- Documentation not created and not kept up-to-date
- doesn’t include security
- not a priority
- Lack of interactive/automated approaches (especially for security)

What is the current state of ML-security?

Highlights from the literature review

Many security challenges for ML exist, such as:

- Attacks more frequent than practitioners realize
- Practitioners lack security awareness and overestimate security of projects
- Functionality valued over security
- security information unavailable
- Lack of penetration testing
- No security documentation standards
- security is costly

The path toward an answer:

Literature review regarding these questions:

- What role does documentation play in security?
- What is the current state of ML-documentation?
- What is the current state of ML-security?
- How can documentation and RE improve ML-security?

Apply the results to ML-documentation:

- Expand existing ML-documentation standards to include security aspects
- Point out challenges and recommendations for future work

How can we improve ML-security through documentation?

Introducing security aspects to existing documentation standards

- Model Cards (Mitchell et al. 2019) – most used model documentation standard
- Datasheets for Datasets (Gebru et al. 2021) – best known dataset documentation standard

Why expand existing standards?

- Better chances of adoption due to lower barrier of entry
- Build on existing peer-reviewed research for a quality solution
- “If it ain‘t broke, don‘t fix it.“

Expanding Model Cards for security

Existing Model Cards standard

- 9 sections
- Security currently not specified in any section
- Some security relevant information present in ethical considerations section

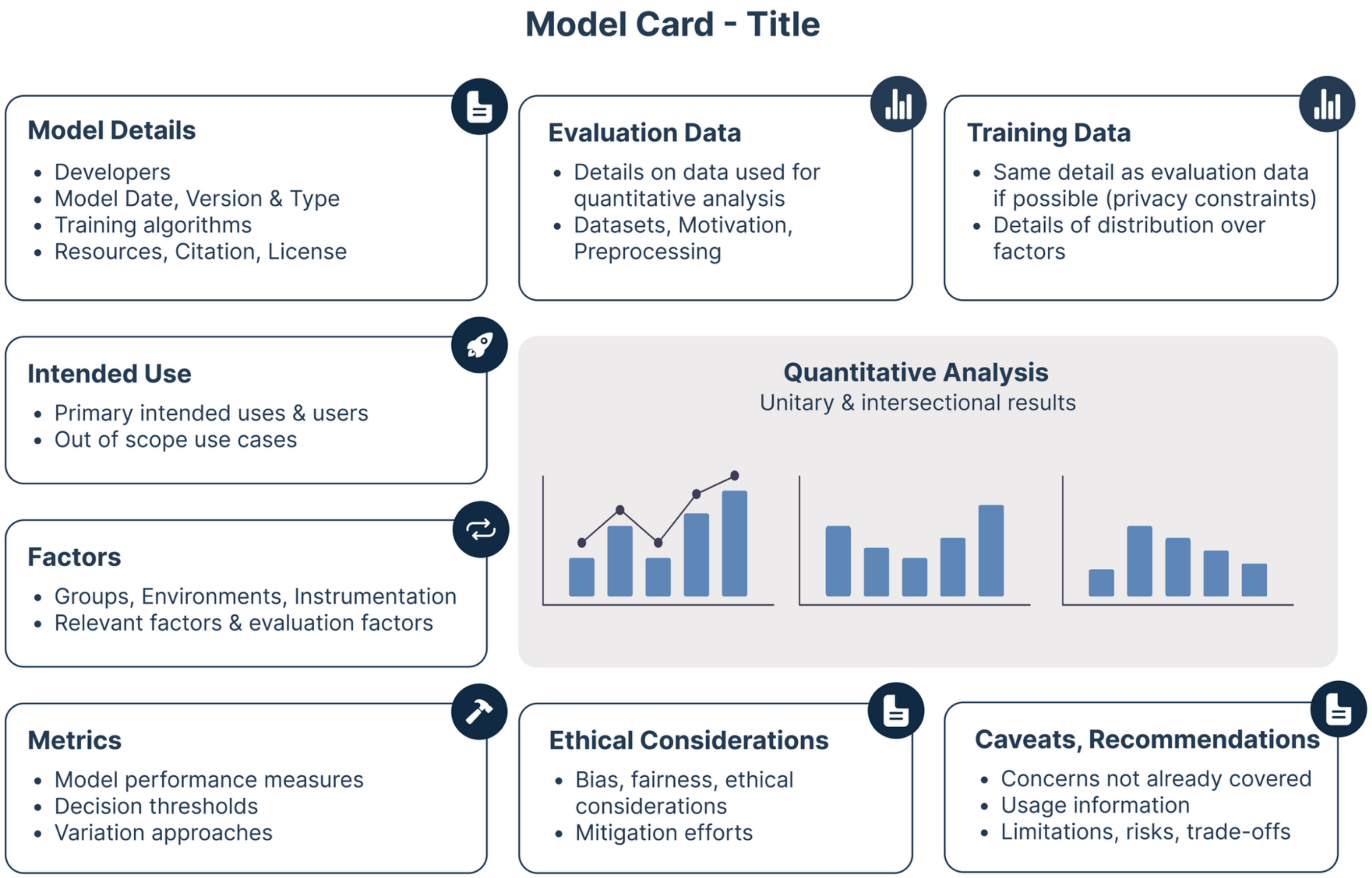

Image Source: trail, https://www.trail-ml.com/blog/ml-model-cards

Proposed new security section:

Risk Analysis	access to model, sensitivity of the data, deployment, model purpose, financial incentive
Training and Evaluation Data	sources, possible data security mitigations
Model Security	model security mitigations, watermarking
Stealing and Inference Attack Mitigations	allowed number of queries, user behavior analysis, user authentification
Security Testing	Security testing that has been done

Expanding Datasheets for Datasets for security

Existing Datasheets for Datasets standard

Motivation

Composition

Collection Process

Preprocessing/
Cleaning/Labeling

Uses

Distribution

Maintenance

- 7 sections containing questions
- Some security relevant information already included:
  - External access to dataset
  - Collection process
  - Sensitive data in dataset
  - Preprocessing of data
  - Maintenance and support

Proposed new security questions

Motivation	Does the purpose for which the dataset was created put it at an increased risk of adversarial attacks?
Preprocessing/Cleaning/Labeling	Were mitigations against adversarial attacks conducted?
Uses	Is there a list of known previous successful attacks on the dataset and any malicious accesses to it?
Maintenance	Will there be security updates to the dataset?

Challenges

Remaining challenges not addressed:

Practitioners not creating/completing documentation

Documentation not being kept up-to-date

Higher effort due to need for model and dataset documentation

Lack of RE in ML-projects

New challenges:

stating security relevant information in accessible documentation

Incomplete/false documentation due to unavailable security information

Lack of RE → unclear security requirements, hindering security documentation

Future Work

- evaluating effectiveness of proposed method
- creating specific security documentation for ML
- increasing security awareness
- adapting Requirements Engineering for ML-security
- applying proposed method to all standards
- creating interactive approaches for the method
- developing unified model and dataset documentation approach

The main question:

How can documentation improve security in ML-systems?

Key Takeaways

- High-quality documentation for ML-systems can significantly aid in increasing security.
- RE in ML-projects can improve both documentation and security.
- documentation and security are often unaddressed in ML-projects.
- existing standards for ML-documentation do not include security
- proposed method of expanding documentation anchors security as part of the documentation process with a low barrier to entry.